

Spatial Distribution of Traumatic Brain Injury Contusions and Post-Traumatic Epilepsy

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Background

- Focal traumatic brain injury (TBI) and contusions are associated with post-traumatic epilepsy (PTE)
- Observed differences confounded by increased injury severity
- Unclear whether focal injury related beyond injury severity, which contusion characteristics (core/edema) associated

Objective: To characterize the spatial distribution of TBI contusions in PTE

Methods

- Transforming Research and Clinical Knowledge in TBI Epileptogenesis Project (TRACK-TBI EPI) data
- Manual contusions segmentation on 2-week T1/FLAIR MRI; masks warped to MNI space
- Disconnection mapping: extend contusion via intersection of normative white-matter tract atlas
- LESYMAP: multivariate contusion map analysis
- Cosine-similarity of stat-map & contusions compared with covariates: age, sex, GCS, total-contusion-volume

Results

Demographics			
Characteristic	non-PTE	PTE	Overall
N	121	14	135
Age, mean (SD)	38.9 (18.4)	33.7 (15.8)	38.4 (18.1)
GCSER, median [IQR]	15.0 [14.0, 15.0]	10.5 [3.0, 15.0]	15.0 [12.0, 15.0]
Female, n (%)	30 (24.8%)	4 (28.6%)	34 (25.2%)

Contusion Mapping Results					
Contusion Characteristic	PTE	non-PTE	Median Difference Cosine Similarity [95% CI]	LESYMAP p-value	p-value
Core + Edema	14	121	0.013 [0.001, 0.031]	0.004	0.036
Core	14	121	0.046 [0.012, 0.096]	0.069	0.010
Edema	10	90	0.026 [-0.008, 0.061]	0.018	0.131
Disconnection	13	129	0.096 [0.038, 0.148]	0.006	0.003

Conclusions

- Contusions burden increased in PTE beyond injury severity - need for external validation
- White matter disconnected by contusions represents strong candidate biomarker
- Contusion-disconnected white matter may carry PTE relevant signal beyond focal injury burden

Results



